

Sai Nihal Konduti

Software Engineer (AI)

+1 (812) 369-9589 | sainihalk2002@gmail.com | sainihalkonduti.com | [LinkedIn](#)

SUMMARY

Backend & AI Engineer (Python/Java, Azure/GCP) building secure microservices and reliable AI systems (RAG, Agents). Scaled a multi-tenant platform for 100+ concurrent tenants, cutting latency by 98% using Azure ETL pipelines. Built Vertex AI workflows to process large datasets efficiently. Microsoft Certified: [Azure Fundamentals \(AZ-900\)](#) | [Azure AI Engineer Associate \(AI-102\)](#)

EDUCATION

M.S. in Computer Science Indiana University, GPA: 3.87/4 Bloomington, IN	Aug 2024 - May 2026
B.Tech. in Computer Science B V Raju Institute of Technology, GPA: 8.75/10 Hyderabad, India	Aug 2020 - Jun 2024

TECH STACK

Languages: Java | Python | TypeScript | SQL | C | JavaScript | R | HTML/CSS
Architectures & Concepts: Distributed Systems | Microservices | RESTful APIs | Postman | ETL Pipelines
AI & Agentic Systems: Generative AI | Large Language Models (LLMs) | RAG | LangChain | Prompt Engineering
Frameworks & Libraries: Spring Boot | Next.js | Node.js | FastAPI
Databases & Data Platforms: MySQL | Azure SQL | MongoDB | Redis | Tableau | Power BI
Cloud & DevOps: Azure | AWS | Google Cloud (Vertex AI) | Docker | Kubernetes | CI/CD (GitHub Actions)
Methodologies: Agile | Scrum

EXPERIENCE

Software Engineer (AI) Infoswift Corp. Irvine, CA Remote	June 2026 – Present
- Engineered a full-stack insurance quote tracking portal to monitor automated Robotic Process Automation (RPA) workflows, providing real-time execution visibility and status tracking across multiple insurance carriers. - Centralized operational visibility for 100+ insurance agents and reduced manual quote status inquiries by 80% by engineering a full-stack RPA tracking portal to monitor real-time automation workflows. - Eliminated client-side token vulnerabilities and secured user sessions against XSS by implementing Backend-For-Frontend architecture using Microsoft Entra External ID and MSAL-Node to manage OAuth flows via encrypted cookies. - Achieved zero-downtime user migration and eliminated administrative overhead for sign-ups by engineering an automated provisioning pipeline in Azure Cosmos DB, dynamically mapping Entra OIDs to legacy agent records to enforce RBAC.	
Research Assistant Indiana University Bloomington	Jan 2026 – May 2026
- Engineered an obstetric triage classification model, achieving a 94.0% Clinical Safety Accuracy and 93.5% F1 Score, by implementing a two-stage supervised fine-tuning pipeline on a Llama 3 8B model. - Minimized hallucinated reasoning and anchored predictions to validated guidelines by developing a gated Retrieval-Augmented Generation (RAG) architecture, dynamically retrieving clinical context via vector search. - Streamlined model training by curating a structured dataset of 788 patient scenarios, utilizing knowledge distillation to translate deterministic decision rules into natural-language training data. - Eliminated cloud API latency and guaranteed local patient data privacy by deploying a 4-bit quantized Small Language Model (SLM) on high-performance computing clusters using Low-Rank Adaptation (LoRA).	
Software Engineer (AI) Infoswift Corp. Irvine, CA Remote	Sep 2025 – Dec 2025
- Built an asynchronous serverless ETL pipeline on Azure Functions using Docker to parallelize Gemini inference at scale, eliminating analytics bottlenecks for 20,000+ files and slashing processing latency by 98% (15m to 30s). - Secured 100% data isolation for 100+ concurrent tenants and prevented cross-tenant leakage by engineering a multi-tenant microservice (FastAPI, Azure). Enforced tenant-scoped paths and HMAC-signed JWTs via GitHub Actions CI/CD. - Reduced API error rates by 40% and maintained 99.9% uptime during traffic spikes by implementing distributed rate-limiting middleware (Redis, Token Bucket). Validated per-tenant quotas using TypeScript and Postman integration tests. - Achieved 90% extraction accuracy for clinical data, transforming reports into structured outputs, by processing PDFs with OCR and engineering a RAG framework (LangChain) to optimize prompt strategies for Gemini Pro.	
AI Engineer Zion Cloud Solutions (ZionAI) Glenview, IL On-site	Jun 2025 – Aug 2025
- Minimized review latency by 30% and enhanced Q&A reasoning by deploying distributed systems on Google Vertex AI (AutoML, Docker, K8s). Benchmarked few-shot prompting across 50+ repositories to streamline policy analysis. - Slashed developer initialization time by 40% by fine-tuning a Small Language Model (Llama-3-8B using LoRA) to automate infrastructure setup. Developed a TypeScript and Next.js interface to streamline the developer onboarding. - Enforced data integrity for 10000+ concurrent records by building a type-safe validation framework to enforce schema consistency across PostgreSQL and Azure SQL. Ensured high availability and compliance during Agile delivery sprints.	

- Accelerated feature deployment cycles by 90% (reducing build times from 20 min to <2 min) by migrating manual tasks to a fully automated CI/CD pipeline using AWS CodePipeline and GitHub Actions for the production environment.
- Cut cloud infrastructure costs by 20% and improved incident response time by 50% by conducting EC2 rightsizing audits, automating S3 archival policies, and engineering an Amazon CloudWatch monitoring suite for 15+ servers.

PROJECTS

CourSelect (Course Advising Platform) [<https://github.com/CJP-Hackathon/Hackathon>]

- Engineered a full-stack AI advising platform using CockroachDB vector embeddings and a 5-tier agentic memory system, achieving sub-3-second semantic searches and enforcing enrollment logic across an 8,983-course catalog.
- Guaranteed zero-downtime reliability via a 4-tier LLM fallback architecture routing between Gemini and local models, while automating session auditing by deploying a Python Flask backend to AWS EC2 and S3.

Hoosier Hub: Full Stack Social App [github.com/IU-HoosierHub/server]

- Engineered a high-concurrency backend scalable to 50,000+ students and 1,000+ clubs, ensuring stability and sub-200ms latency, by architecting APIs (OpenAPI) and benchmarking MongoDB schemas.
- Established a production-grade DevOps environment by building a fully automated CI/CD pipeline with Docker and GitHub Actions, while securing the platform with JWT authentication using Spring Boot.

Distributed Serverless Log Processing System with Azure [<https://github.com/nihal2405/CloudLogProcessing>]

- Processed 10GB+ of daily raw logs in real-time by building a high-throughput ingestion engine using Azure Functions to parse, split, and structure unstructured data streams, eliminating 90% of manual parsing overhead.
- Reduced query time for system audits from hours to seconds by structuring unstructured text into queryable insights, using a parallel parsing engine to extract timestamps and severity levels into Azure SQL via optimized bulk inserts.

The Untold Stories of IPL: Advanced EDA & Visualization [<https://github.com/nihal2405/IPL-STORY>]

- Conducted advanced Exploratory Data Analysis on 16 years of IPL match data from 2008 to 2024 using Power BI and Tableau to design complex visualizations like Rose Charts and Violin Plots that map player consistency.
- Analyzed the correlation between competitive win percentages and commercial brand value to identify key statistical drivers of franchise popularity beyond title counts and transformed aggregate stats into compelling data narratives.

PUBLICATION

[Enhancing Soft Skill Development with ChatGPT and VR. IEEE International Conference - RMKMATE 2023](#)